# ECON 150 · Lecture 2
# Elasticity

How strongly quantity responds to price and income

# Price elasticity of demand

$E_d$ = %ΔQ / %ΔP (report the absolute value)

```
Midpoint method: %ΔQ = (Q2 − Q1) / ((Q1 + Q2)/2)
```

# Midpoint example

```
P: 4 → 6, Q: 120 → 80
%ΔQ = −40 / 100 = −40%
%ΔP = 2 / 5 = 40%
|E| = 1 (unit elastic)
```

# Elastic or inelastic?

- $|E| > 1$ elastic, $|E| < 1$ inelastic, $|E| = 1$ unit elastic
- More elastic with: close substitutes, luxuries, longer time horizon, large budget share, narrow market definition

# Total revenue test

- Elastic: price $\downarrow \rightarrow$ total revenue $\uparrow$
- Inelastic: price $\uparrow \rightarrow$ total revenue $\uparrow$
- Unit elastic: total revenue unchanged

# Income and cross-price elasticity

- Income elasticity $> 0$ normal, $< 0$ inferior, $> 1$ luxury
- Cross-price elasticity $> 0$ substitutes, $< 0$ complements

# Elasticity of supply

- Es = %ΔQs / %ΔP
- Supply is more elastic in the long run than the short run

# Tax incidence

- A tax splits between buyers and sellers
- The more **inelastic** side bears more of the tax
- Who legally pays does not decide who bears it